

Assignment – 1

80/100

Question: 1

1. Graphene is a two-dimensional (2D) crystal with carbon atoms arranged in a honeycomb structure as shown in Figure 1. Using the above information and Figure 1, answer the following questions.

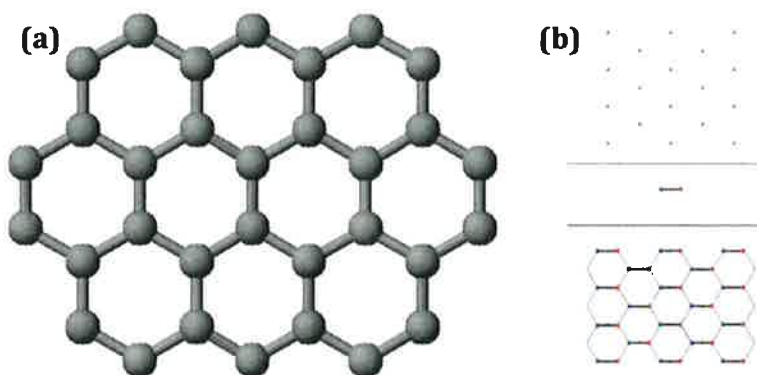
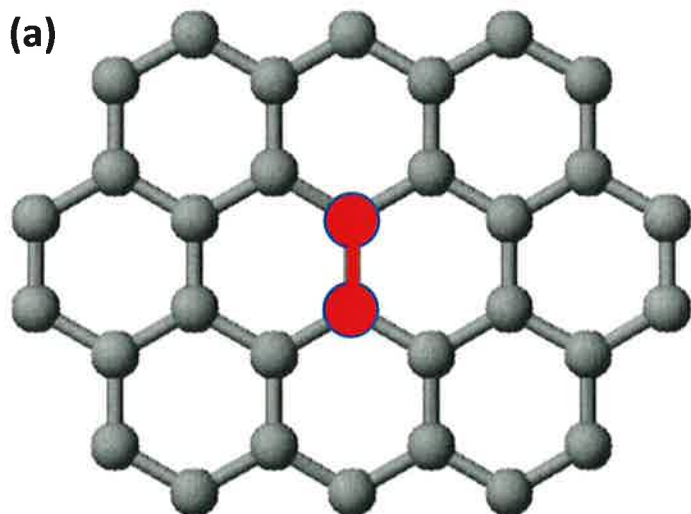


Figure 1: Graphene Lattice.

- (a) Assuming that the unit cell is a hexagon of carbon atoms (Figure 1 (a)), how many atoms of carbon are there in each unit cell (Explain)?
(b) Crystal Structure = Lattice + Basis. In Figure 1 (b), designate: (i) lattice and its type, (ii) basis and how many atoms in basis, and crystal structure.

Answer:



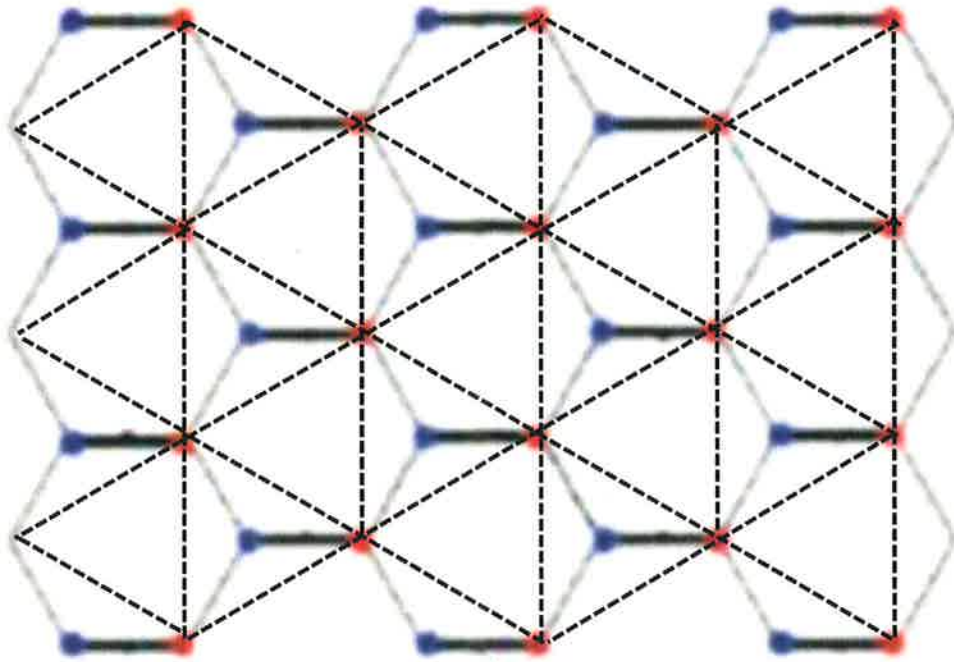
Calculations:

There are total 6 carbon atoms in independent hexagonal unit cell. In case of graphene lattice two carbon atoms from each unit cell are being shared.

$$6 \times \left(\frac{1}{3}\right) = 2$$

So, each unit cell has 2 carbon atoms.

(b)



Single 2D layer of graphene consists of honeycomb like hexagonal closed pack (HCP) crystal structure.

Sub lattice spacings are: $2 \cdot \sin 60^\circ \cdot a = \sqrt{3} a$

Bravais sublattice is triangular. Basis consists of 2 carbon atoms. 

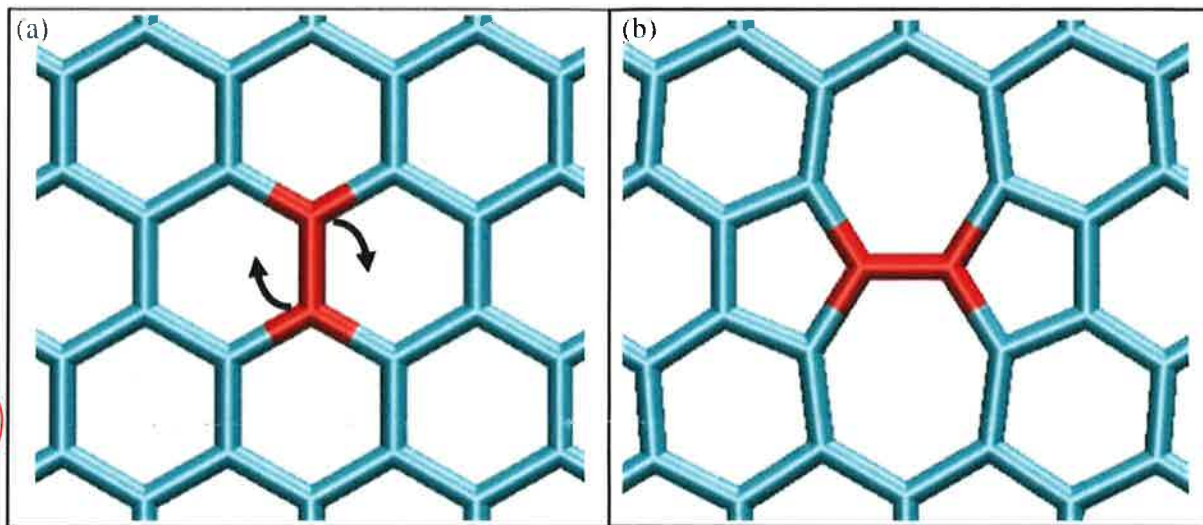
Question:2

2. What is Stone-Wales (SW) defect in graphene and state SW defect's consequences on the properties of graphene?

Answer:

(Reference: Ma, Jie, et al. "Stone-Wales defects in graphene and other planar sp^2 -bonded materials." *Physical Review B* 80.3 (2009): 033407.)

Stone-Wales defect:



Stone-Wales defect (a point defect) involves an in-plane 90° rotation of two carbon atoms with respect to the midpoint of the bond. (as represented in the above schematic). As a result, 4 hexagons of regular graphene structure are converted into two heptagons and 2 pentagons.

(Reference: Kang, Joongoo, et al. "Effect of atomic-scale defects on the low-energy electronic structure of graphene: Perturbation theory and local-density-functional calculations." *Physical Review B* 77.11 (2008): 115453.)

Stone-Wales (SW) defects in graphene structure are responsible for fracture and embrittlement of carbon nanostructures. SW defects are responsible for altering electronic properties (band structure and density of states) of graphene, which apparently induce changes in chemical reactivity towards adsorbates.

(Reference: Safarian, Sobhan, and Masoud Tahani. "Study of Stone-wales Defect on Elastic Properties of Single-layer Graphene Sheets by an Atomistic based Finite Element Model." *International Journal of Engineering* 31.3 (2018): 456-463.)

Sine wave and cosine wave like structural topological deformation occurs by Stone-Wales defect. Young's modulus and shear modulus in both directions change as a result of it. Approximately 4-5% decrease in the Young's modulus is observed at SW defect site in comparison with defect free sites along zig-zag and arm-chair directions. Also, shear modulus decreases approx. 4-9%.

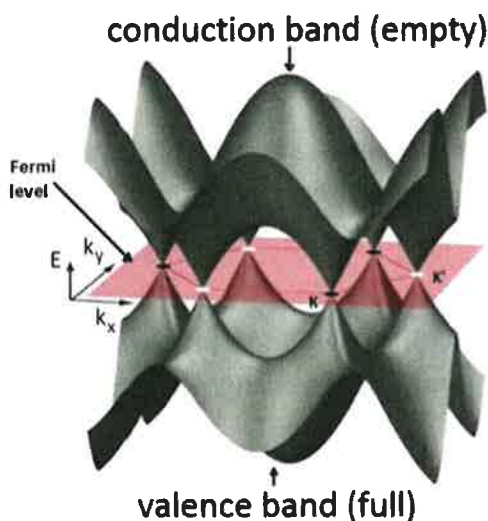
(Reference: Yin, HengChuang, et al. "Stone-Wales graphene: A two-dimensional carbon semimetal with magic stability." *Physical Review B* 99.4 (2019): 041405.)

SW defect induced altered electronic structure induces semi-metal properties in graphene with distorted and strongly anisotropic Dirac cones. But, remarkable energy stability can be observed at SW defect points.

Question:3

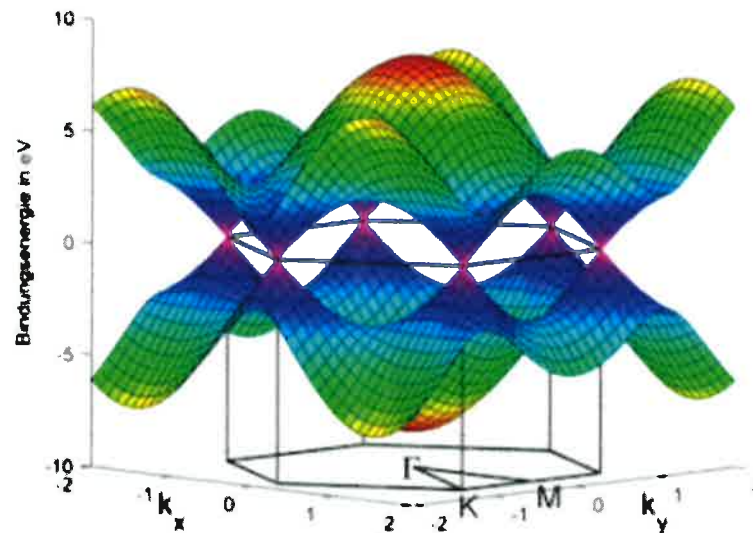
3. Intrinsic monolayer graphene is a zero bandgap ($E_g = 0$ eV) semiconductor. Show the band structure (Energy-Momentum Linear Dispersion) of an intrinsic monolayer graphene. Suggest some techniques (minimum two) to open bandgap in graphene.

(Reference: Zhang, Xiang, et al. "Graphene's potential in materials science and engineering." *RSC Advances* 4.55 (2014): 28987-29011.)



Dirac cone arrangement electronic band structure in graphene

The Fermi structure of graphene is characterized by 6 double-cones. In the pristine form, the Fermi level is located at the points connecting valence and conduction bands, where density of states will be zero. The Fermi level can be improved with by applying electric field.



b) graphene π - and π^* -band structure over the complete Brillouin zone

(Reference: <https://www.uni-muenster.de/Physik.PI/Zacharias/en/research/graphene/graphene.html>)

The π and π^* bands in graphene connect at a single point at the Fermi level. (K – point in the Brillouin zone. The electrons with such momentum the dispersion is linear.

Band gap opening in graphene:

(**Reference:** Jariwala, Deep, Anchal Srivastava, and Pulickel M. Ajayan. "Graphene synthesis and band gap opening." *Journal of nanoscience and nanotechnology* 11.8 (2011): 6621-6641.)

Method – 1: Substrate induced band gap opening

Epitaxial single layers of h-BN followed by graphene were synthesized on Ni (111). The graphene/h-BN/Ni(111) system showed a bandgap of 0.5 eV. Another theoretical study shows if the graphene is deposited on oxygen free SiO_2 surfaces can induce band gap of 0.52 eV. It occurs if the substrate is hydrogenated i.e. free from OH radicals can avoid covalent interaction between graphene and oxygen and hence metallic properties of graphene are preserved.

Method – 2: Chemical substitution doping

Nitrogen doping can convert graphene into p-type semiconductor. Experimentally source of ammonia gas mixed methane is used for CVD synthesis in order to achieve nitrogen doping. Boron doping is performed in the similar manner to achieve p-type semiconductor attributes to graphene.

Question:4

Calculate lattice vectors a_1 and a_2

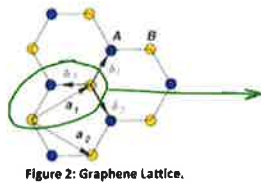
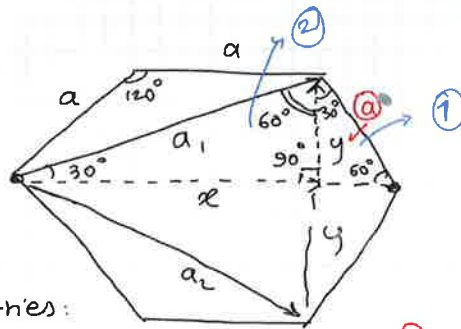


Figure 2: Graphene Lattice.



Ans from simple trigonometry:

$$a_1 = 2y \quad \therefore y = \frac{\sqrt{3}}{2} a \quad (\text{i.e. } \sin 60^\circ)$$

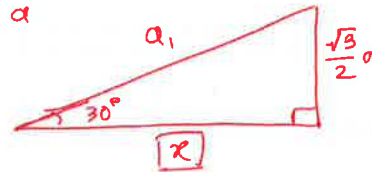
$$a_2 = a_1 = 2y \quad \dots \text{absolute value}$$

by vector-properties:

$$a_1 = x + y \rightarrow \text{I}$$

$$a_2 = x - y \rightarrow \text{II}$$

y in terms of a
 $y = \frac{\sqrt{3}}{2} a$



$$\frac{x}{a_1} = \cos 30^\circ$$

$$x = a_1 \cos 30^\circ = \frac{\sqrt{3}}{2} a_1$$

Hence, $a_1 = x + y$

$$\therefore \vec{a}_1 = \frac{\sqrt{3}}{2} \vec{a}_1 + \frac{\sqrt{3}}{2} a$$

$$\vec{a}_1 \left(1 - \frac{\sqrt{3}}{2} \right) = \frac{\sqrt{3}}{2} a$$

$$\therefore \vec{a}_1 \left(\frac{2 - \sqrt{3}}{2} \right) = \frac{\sqrt{3}}{2} a$$

$$\therefore \boxed{\vec{a}_1 = \frac{\sqrt{3}}{(2 - \sqrt{3})} a} \dots \text{Ans}$$

$$a_2 = x - y$$

$$\vec{a}_2 = \frac{\sqrt{3}}{2} \vec{a}_2 - \frac{\sqrt{3}}{2} a$$

$$\vec{a}_2 \left(1 - \frac{\sqrt{3}}{2} \right) = -\frac{\sqrt{3}}{2} a$$

$$\vec{a}_2 \left(\frac{2 - \sqrt{3}}{2} \right) = -\frac{\sqrt{3}}{2} a$$

$$\therefore \boxed{\vec{a}_2 = \frac{\sqrt{3}}{(\sqrt{3} - 2)} a} \dots \text{Ans}$$